Muhammad Talal Ahmed

Software Engineer | Python · FastAPI · Full-Stack · Automation

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Computer Science graduate (CGPA 3.79, Bronze Medal) and software engineer building dependable web applications, backend services, automation tools, and AI-powered platforms. Comfortable moving from an idea to an implementation across Python, FastAPI, PostgreSQL, and modern frontend stacks, with a security-minded approach throughout development.

TECHNICAL SKILLS

Languages	Python, JavaScript/TypeScript, SQL, C++
Frameworks	FastAPI, React, Next.js, Node.js, Astro, Hugo
Data & Automation	PostgreSQL, MongoDB, ChromaDB, ETL pipelines, PDF/data extraction, n8n workflows
Tools	Git, GitHub, Docker, Linux, Prisma ORM, REST APIs, server-sent events
Also:	LangGraph, RAG, Scikit-learn, Odoo ERP, Wazuh, pfSense, GRC risk assessment

PROJECTS

AI-Driven SRS Generator (Final Year Project)

Python, FastAPI, LangGraph, Next.js, PostgreSQL, ChromaDB

- Built a full-stack AI-powered platform that generates IEEE 830-compliant software requirements specifications through a guided multi-agent workflow.
- Orchestrated agents in LangGraph with structured model outputs and PostgreSQL-backed state, so long requirement sessions survive restarts.
- Implemented retrieval-augmented generation over a curated technical knowledge base in ChromaDB to ground generated requirements in source material.
- Connected a FastAPI backend to a Next.js frontend over REST and server-sent events for streaming generation.

ML for Malware Detection

Python, Scikit-learn, Pandas

- Built a Python ML project for polymorphic malware detection using machine learning techniques for security-focused binary classification.

Timetable Parser

Python, PDF processing, JSON

- Developed a Python-based PDF parsing pipeline to extract timetable data from unstructured university documents and emit normalized JSON.

Department Website

Astro, TypeScript

- Developed the official university CS department website with Astro, with reusable components and fast page loads.

EXPERIENCE

SOC Analyst Intern — ITSOLERA PVT LTD 01/2026 – 04/2026

SOC Internship Programme · Team Lambda · 6 assessed tasks

- Built a Wazuh-to-n8n workflow that parses, filters, and forwards alerts, delivering only actionable events instead of raw dashboard noise.
- Integrated a software firewall with Wazuh and wrote rules to block country traffic, restrict access to specified sites, and enforce administrator privileges.

IT Intern — Jauharabad Sugar Mills Pvt Ltd 07/2025 – 09/2025

- Provided tier-1 support for Windows, hardware, and network faults in a production facility, and validated roughly 9,200 QR-coded records.

WRITING & OPEN SOURCE

40+ public repositories at github.com/Ta1al across TypeScript/Node tooling, Python services, Astro and Hugo sites, and Discord bots.
Published technical writeups at ta1al.com/blog, including AI security and prompt injection work.

EDUCATION

BS Computer Science — Sargodha University 2022 – 2026

CGPA 3.79 (Bronze Medal) · President, Programming Society

CERTIFICATIONS

National Skills Competency Test — Rank 65 nationwide (Top 0.2%)
Python for Everybody Specialization (Coursera)